

1	A signal $y[n]$ is an up-sampled version of $x[n]$ by a factor of 2. $y[n] = \begin{cases} x\left[\frac{n}{2}\right] & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases}$ If N is the time period of $y[n]$, determine the time period of $x[n]$ based on N being even or odd.	[2 Marks]
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Q1

$$y(n) = \begin{cases} x\left[\frac{n}{2}\right] & n \rightarrow \text{even} \\ 0 & n \rightarrow \text{odd} \end{cases}$$

$N \rightarrow \text{even}$.

$$x(n+N_1) = x(n)$$

$\underline{x(n+N_1)}$ $\rightarrow$ $\underline{y(2(n+N_1))}$ $\underline{n+N_1} \rightarrow \text{even}$

$$= y(2n+2N_1) = y(2n) \quad \underline{n \rightarrow \text{even}}$$

$\underline{N_1 = \frac{N}{2}}$ $\underline{n+N_1} \rightarrow \text{odd}$

$N \rightarrow \text{odd}$ $\underline{y(2n)} = \underline{x(n)}$

$$y(2n+2N_1) = y(2n+N+N) = y(2n) = x(n)$$

2

(a) Given $x(t) = 3u(t) + tu(t) - (t-1)u(t-1) - 5u(t-2)$, sketch the signal with clearly labelling of its values. [2 Marks]

(b) Derive the expression $g(t)$ in Fig. 1 using t and $u(t)$.

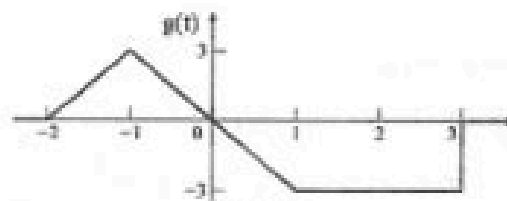


Fig. 1.

B2

(a) $z(t) = 3u(t) + tu(t) - (t-1)u(t-1) - 5u(t-2)$

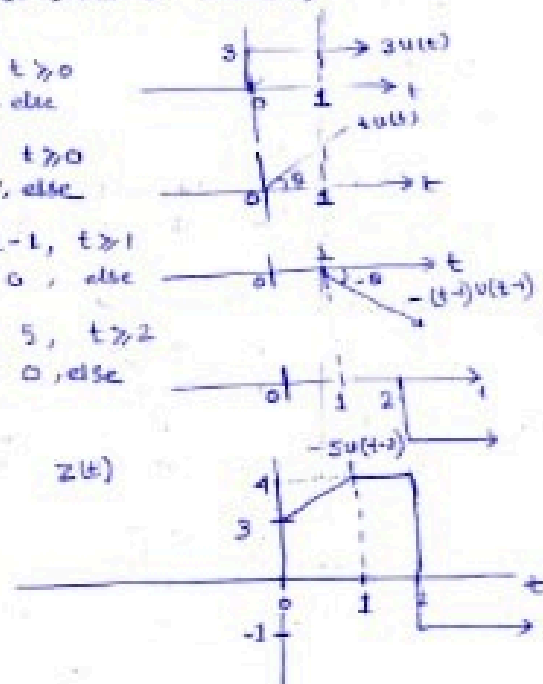
Details:

$$3u(t) = \begin{cases} 3, & t \geq 0 \\ 0, & \text{else} \end{cases}$$

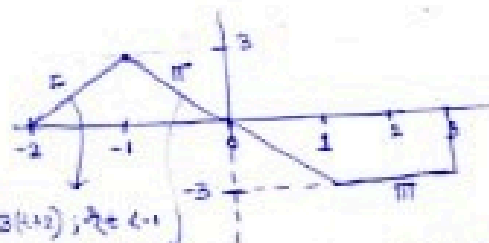
$$tu(t) = \begin{cases} t, & t \geq 0 \\ 0, & \text{else} \end{cases}$$

$$(t-1)u(t) = \begin{cases} t-1, & t \geq 1 \\ 0, & \text{else} \end{cases}$$

$$5u(t-2) = \begin{cases} 5, & t \geq 2 \\ 0, & \text{else} \end{cases}$$



(b)



Details:

$$\text{I } g(t) = 3(t+2), \quad t < -1$$

$$\text{II } g(t) = -3t, \quad -1 < t < 1$$

$$\text{III } g(t) = -3, \quad 1 < t < 3$$

$$g(t) = 0, \quad t > 3$$

$$g(t) = \text{I} + \text{II} + \text{III}$$

$$= 3(t+2)[u(t+2) - u(t+1)] - 3t[u(t+1) - u(t-1)] + 3[u(t-1) - u(t-3)]$$

$$= 3(t+2)u(t+2) - 3(t+2)u(t+1) - 3tu(t+1) + 3tu(t-1) - 3u(t-1) + 3u(t-3)$$

$$= 3(t+2)u(t+2) - u(t+1)[3t+6+3t] + u(t-1)[3t-3] + 3u(t-3)$$

$$= 3(t+2)u(t+2) - 6(t+1)u(t+1) + 3(t-1)u(t-1) + 3u(t-3)$$

3	$x[n] = \cos\left[\pi - \frac{n\pi}{6}\right] \quad -2 \leq n \leq 2$, determine the signal is energy signal or power signal or neither of these. Justify your answer with suitable numerical calculations.	[2 Marks]
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$$\begin{aligned}
 x[n] &= \cos\left[\pi - \frac{n\pi}{6}\right] \quad -2 \leq n \leq 2 \\
 &\text{Energy signal or power signal.} \\
 E &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\
 &= \sum_{n=-2}^2 \cos^2\left[\pi - \frac{n\pi}{6}\right] \\
 &= \sum_{n=-2}^2 \left[\frac{1}{2} + \frac{1}{2} \cos\left[2\pi - \frac{n\pi}{3}\right] \right] \quad \cos^2\theta = \frac{1 + \cos 2\theta}{2} \\
 &= \sum_{n=-2}^2 \left[\frac{1}{2} + \frac{1}{2} \cos\left(\frac{n\pi}{3}\right) \right] \\
 &= \frac{5}{2} + \underbrace{\frac{1}{2} \cos\left(\frac{2\pi}{3}\right)}_{\substack{(n=2) \\ (n=-2)}} + \underbrace{\frac{1}{2} \cos\left(\frac{\pi}{3}\right)}_{\substack{(n=1) \\ (n=-1)}} + \frac{1}{2} \cos(0)_{(n=0)} \\
 &\Rightarrow \frac{5}{2} + \cos\left(\frac{2\pi}{3}\right) + \cos\left(\frac{\pi}{3}\right) + \frac{1}{2} = 3 + 0.5 - 0.5 \\
 &= \underline{\underline{3 \text{ units}}} \\
 &\underline{\underline{\text{Energy signal}}}
 \end{aligned}$$

4	Check if the systems given below are linear and time invariant	[3 Marks]
	(i) $y[n] = \alpha x[1 - n]$	
	(ii) $y[n] = x[n] + \frac{2}{x[n-1]}$	
	(iii) $y[n] - y[n-1] = \alpha x[n] - \beta x[n-1]$	

(1)

$$y(n) = \alpha x(1-n)$$

$$x_1 \rightarrow y_1 \quad x_2 \rightarrow y_2$$

$$x(n) = \alpha x_1(n) + \beta x_2(n)$$

$$y(n) = \alpha y_1(n) + \beta y_2(n)$$

$$T[x(n)] = \alpha [\alpha x_1(n) + \beta x_2(n)]$$

$$= \alpha [\alpha x_1(1-n) + \beta x_2(1-n)]$$

$$= \alpha y_1(n) + \beta y_2(n)$$

$$= y(n)$$

0.5 / Linearity Yes

$$T[x(n-k)] = \alpha x(1-n-k)$$

$$y(n-k) = \alpha x(1-(n-k)) = \alpha x(1-n+k)$$

$$\therefore T[x(n-k)] \neq y(n-k)$$

0.5 / Time varying

$$(ii) \quad y(n) = x(n) + \frac{2}{x(n-1)}.$$

Linearity.

$$T(x(n)) = T[a x_1(n) + b x_2(n)]$$

$$= a x_1(n) + b x_2(n) + \frac{2}{a x_1(n-1) + b x_2(n-1)}.$$

$$y(n) = a y_1(n) + b y_2(n)$$

$$= a \left[x_1(n) + \frac{2}{x_1(n-1)} \right] + b \left[x_2(n) + \frac{2}{x_2(n-1)} \right]$$

$$= a x_1(n) + b x_2(n) + 2 \cdot \left[\frac{a}{x_1(n-1)} + \frac{b}{x_2(n-1)} \right]$$

∴ Not linear as $y(n) \neq T[x(n)]$

Time invariance

$$T[x(n-k)] = x(n-k) + \frac{2}{x(n-k-1)}.$$

$$y(n-k) = x(n-k) + \frac{2}{x(n-k-1)}$$

Time invariance — Yes.

$$(iii) \quad y(n) - y(n-1) = \alpha x(n) - \beta x(n-1).$$

linearity. $x_1(n) \rightarrow y_1(n)$
 $x_2(n) \rightarrow y_2(n).$

$$x(n) = a x_1(n) + b x_2(n); \quad y(n) = a y_1(n) + b y_2(n).$$

$$y_1(n) - y_1(n-1) = \alpha x_1(n) - \beta x_1(n-1)$$

$$y_2(n) - y_2(n-1) = \alpha x_2(n) - \beta x_2(n-1)$$

$$a [y_1(n) - y_1(n-1)] = a (\alpha x_1(n) - \beta x_1(n-1)).$$

$$+ b [y_2(n) - y_2(n-1)] = b (\alpha x_2(n) - \beta x_2(n-1))$$

$$\Rightarrow a [y_1(n) + b [y_2(n)] - [a y_1(n-1) + b y_2(n-1)]]$$

$$- a [\alpha x_1(n) + b x_2(n)] + \beta [\alpha x_1(n-1) + b x_2(n-1)] = 0$$

$$\Rightarrow y(n) - y(n-1) = \alpha x(n) - \beta x(n-1) = 0.$$

Linearity Yes.

Time invariance

$$\cancel{T(x(n-k))} = \cancel{y_1(0)}$$

$$y(n) = y(n-1) + \alpha x(n) - \beta x(n-1)$$

shift i/p by k

$$y_1(n) = y_1(n-1) + \alpha x(n-k) - \beta x(n-k-1)$$

shift o/p by k

$$y(n-k) = y(n-k-1) + \alpha x(n-k) - \beta x(n-k-1)$$

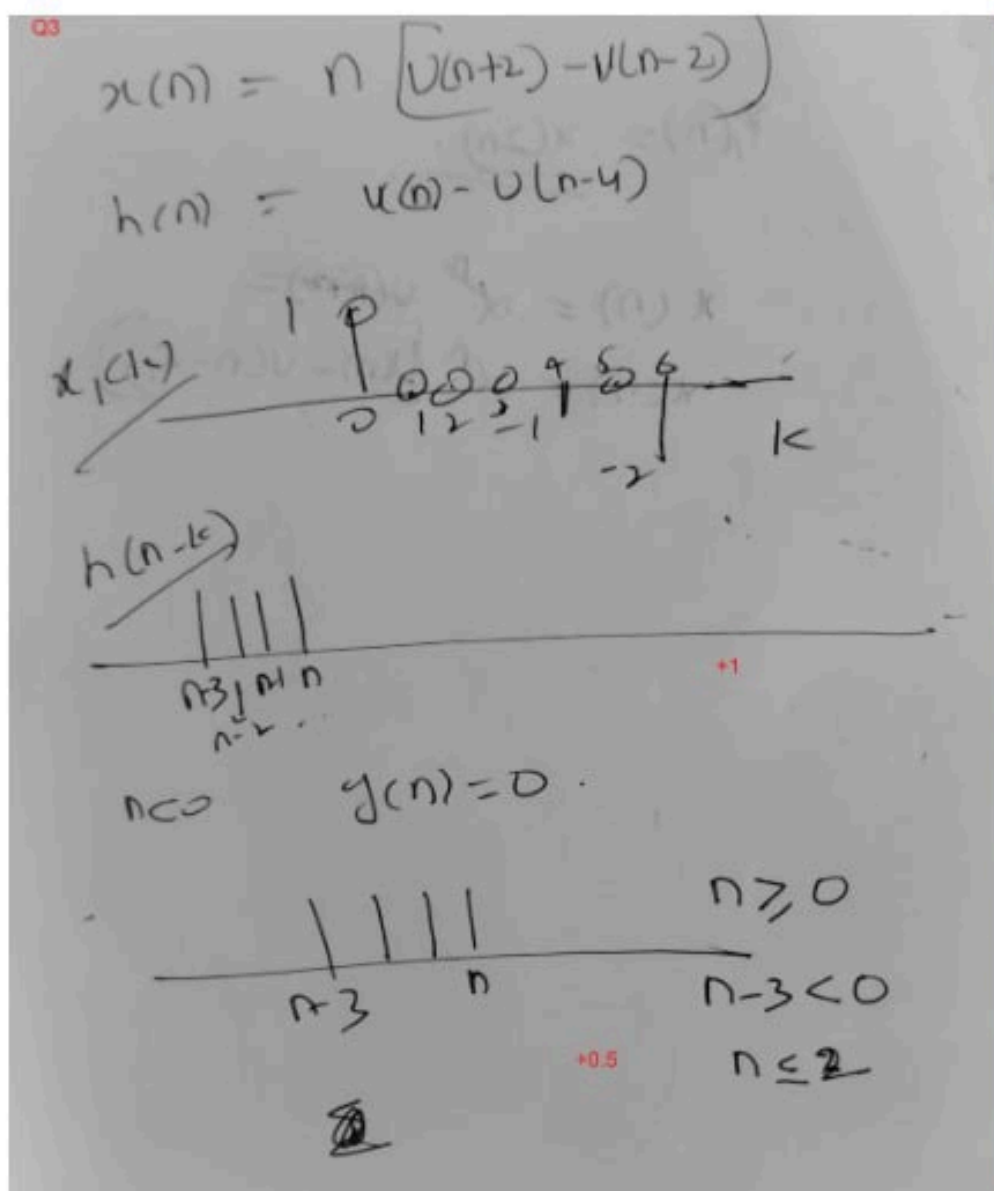
$$\therefore y_1(n) = y(n-k) \rightarrow \text{o/p are same.}$$

Hence

Time invariant

to 5

5	if $x[n] = n(u[n+2] - u[n-2])$ and $h[n] = (u[n] - u[n-4])$ Determine the following convolution sum $y[n] = x \left[1 - \frac{n}{2}\right] * h[n]$ And plot the signals $x[n]$ and $h[n]$ and the convolution output $y[n]$. [DO NOT use the tabular method for convolution].	[4 Marks]
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$$x_1(n) = \{1, 0, 0, 0, -1, 0, -2\}$$

$$h(n) = \{1, 1, 1, 1\}$$

$$y(0) = \sum_{k=0}^6 x_1(k) h(-k)$$

$$= 1 \cdot 1 = 1$$

$$y(1) = \sum_{k=0}^6 x_1(k) h(1-k)$$

$$= 1 \cdot 1 + 0 \cdot 1 = 1$$

$$y(2) = \sum_{k=0}^6 x_1(k) h(2-k)$$

$$= 1 \cdot 1 + 0 + 0 = 1$$

$$y(3) = 1 + 0 + 0 + 0 = 1$$

$$y(4) = 1 \cdot 0 + 0 + 0 + 0 + (-1) = -1$$

$$y(5) = 0 + 0 + 0 + 0 + (-1) \cdot 1 = -1$$

$$y(6) = 0 + 0 + 0 + 0 + (-1) \cdot 1 + 0 + (-2) \cdot 1 = -3$$

$$y(7) = 0 + 0 + 0 + 0 + (-1) \cdot 1 + 0 + (-2) \cdot 1 = -3$$

$$y(8) = 0 + 0 + 0 + 0 + (-1) \cdot 1 + 0 + (-2) \cdot 1 = -2$$

+2.5

$$y(n) = \{1, 1, 1, 1, -1, -1, -3, -3, -2\}$$

Suppose an LTI system produces a continuous-time signal $x(t)$ as shown in Fig. 2. Decompose the signal into its even and odd components (using graphical representation only) with detailed labelling of the values.

[5 marks]

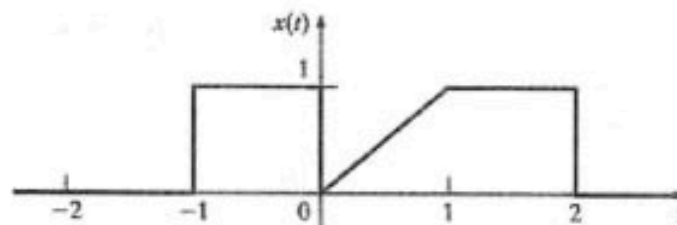
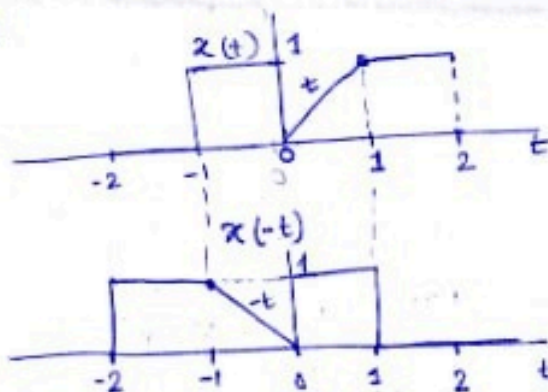
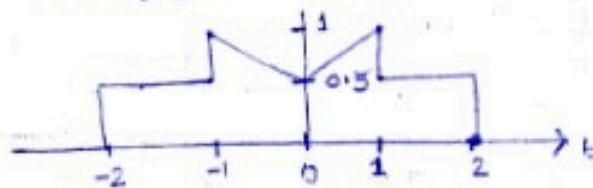


Fig. 2.

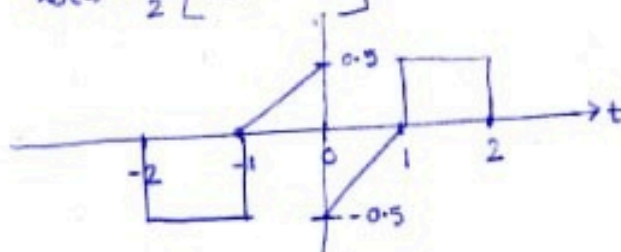
Q6



$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$



$$x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$



Details:

$$-2 < t < -1 \Rightarrow 0.5$$

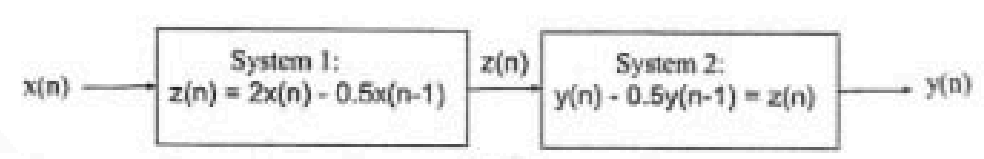
$$-1 < t < 0 \Rightarrow \frac{1-t}{2}$$

$$0 < t < 1 \Rightarrow \frac{1+t}{2}$$

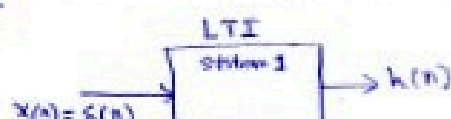
$$-2 < t < -1 \Rightarrow -0.5$$

$$-1 < t < 0 \Rightarrow \frac{1}{2} [1+t]$$

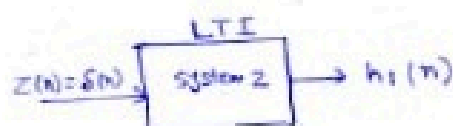
$$0 < t < 1 \Rightarrow \frac{1}{2} [t-1]$$

7	Suppose the input-output relationship of two cascaded LTI systems is shown in Fig. 3. Justify whether System 2 is the inverse of System 1, providing detailed reasoning. Assume that System 2 is initially in the relaxed condition.  Fig. 3.	[3 Marks]
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$$h_1(n) = 2\delta(n) - 0.5\delta(n-1) \quad \text{--- (1)}$$



$$h_2(n) - 0.5h_2(n-1) = \delta(n) \quad \text{--- (2)}$$

or

$$y(n) - 0.5y(n-1) = \delta(n); \quad y(n) = h_2(n)$$

$$n=0, \quad y(0) - 0.5y(-1) = \delta(0) = 1$$

As system is initially in relaxed condition $y(n)=0, n<0$

$$\Rightarrow y(0) = 1 = \left(\frac{1}{2}\right)^0$$

$$n=1, \quad y(1) - 0.5y(0) = \delta(1) = 0$$

$$\Rightarrow y(1) = 0.5 \times 1 + 0 = \frac{1}{2} = \left(\frac{1}{2}\right)^1$$

$$n=2, \quad y(2) - 0.5y(1) = \delta(2) = 0$$

$$\Rightarrow y(2) = 0.5 \times \frac{1}{2} + 0 = \left(\frac{1}{2}\right)^2$$

$$\vdots$$

$$y(n) = \left(\frac{1}{2}\right)^n$$

$$\therefore h_2(n) = \left(\frac{1}{2}\right)^n u(n)$$

If system 2 is inverse of 1,

then $h(n) \otimes h_2(n) = \delta(n)$

L.H.S

$$[2\delta(n) - 0.5\delta(n-1)] \otimes h_2(n)$$

$$= 2\delta(n) \otimes h_2(n) - 0.5\delta(n-1) \otimes h_2(n)$$

$$= 2h_2(n) - 0.5h_2(n-1) \quad ; \quad \text{as } x(n) \otimes \delta(n) = x(n)$$

$$x(n) \otimes \delta(n-1) = x(n-1)$$

$$= 2\left(\frac{1}{2}\right)^n u(n) - \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

$$= \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(n-1)$$

$$= \left(\frac{1}{2}\right)^n [u(n) - u(n-1)]$$

$$= \left(\frac{1}{2}\right)^n [2u(n) - u(n-1)] \neq \delta(n)$$

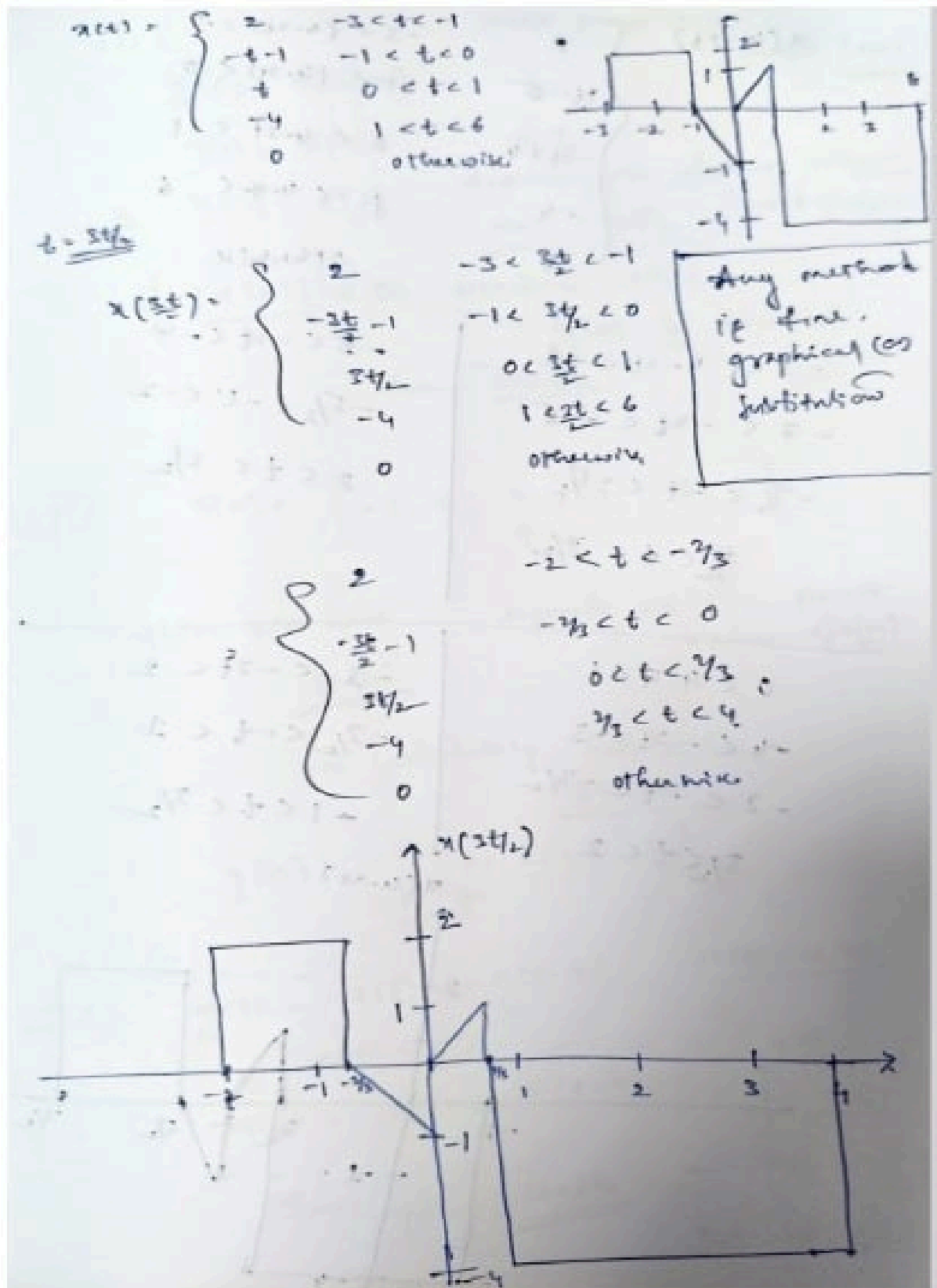
$\therefore$ system 2 is not inverse of 1.

8

$$\text{If } x(t) = \begin{cases} 2 & -3 < t < -1 \\ -t-1 & -1 < t < 0 \\ t & 0 < t < 1 \\ -4 & 1 < t < 6 \\ 0 & \text{Otherwise} \end{cases}$$

[3 Marks]

Determine (i) $x\left(\frac{3}{2}t\right)$ and (ii) $x(4-2t)$ and plot both signals.



$x(4-2t)$

$$= \begin{cases} 2 & -3 < (4-2t) < -1 \\ 2t-5 & -1 < (4-2t) < 0 \\ -2t+4 & 0 < 4-2t < 1 \\ -4 & 1 < 4-2t < 6 \\ 0 & \text{otherwise} \end{cases}$$

$$-3 < 4-2t < -1$$

$$-7 < -2t < -5$$

$$-7/2 < -t < -5/2$$

$$5/2 < t < 7/2$$

$$-5 < -2t < -4$$

$$-5/2 < -t < -2$$

$$2 < t < 5/2$$

$$0 < 4-t < 1$$

$$-4 < -t < -3$$

$$-2 < -t < -3/2$$

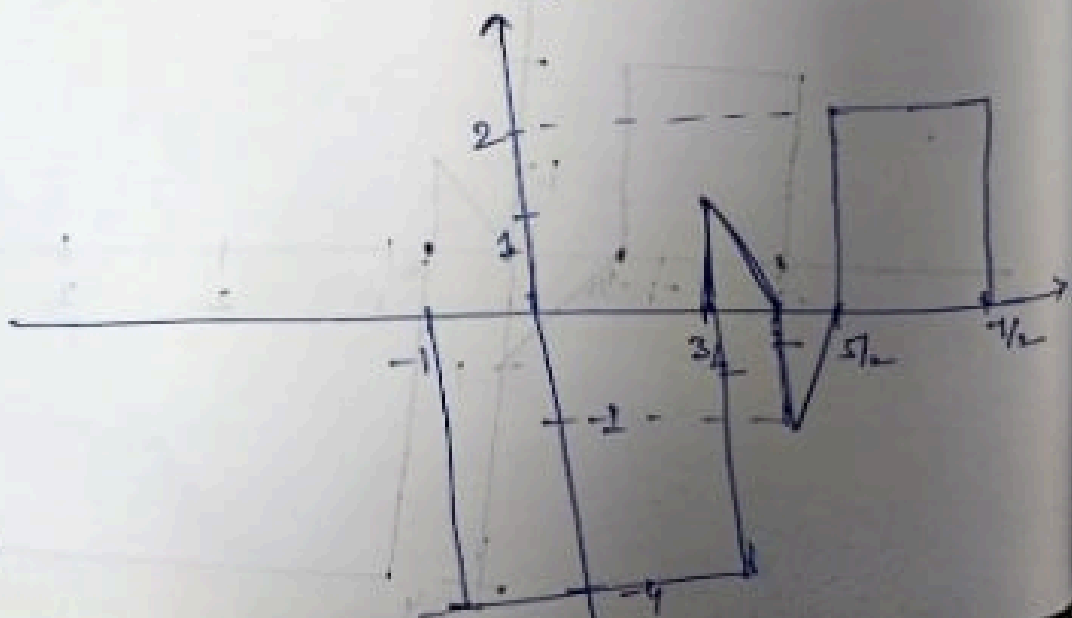
$$3/2 < t < 2$$

$$-3 < -2t < 2$$

$$-3/2 < -t < 1$$

$$-1 < t < 3/2$$

$x(4-2t)$



9

Determine whether the following systems are static/Memoryless or not static/memory, stable or not stable. Justify your answer with suitable numerical calculations.

[3 Marks]

- (i) $y[n] = x[\delta[n]]$ where $\dots x[-1], x[0], x[1], \dots$ all are bounded in magnitude.
 (ii) $y[n] = x[n]u[n]$ where $x[n] = \{-1, 4, \frac{2}{n}, 0, -5, 1\}$
 (iii) $y(t) = x(t-1) - x(t-4)$ where $x(t) = u(t)$

(i)

$$y[n] = x[\delta[n]]$$

$$y[0] = x[1]$$

$$n=1 \quad y[1] = x[0]$$

Static / Not Static

stable / not stable.

 $n=0$

depending on future values,
 past values
Not static.

if $|x[n]| < B$ bounded then
 $y[n]$ is bounded.

then stable.

(ii) $x[n] = \{ -1, 4, \underset{\uparrow}{2}, 0, -5, 1 \}$

$y[n] = x[n] u[n]$ depends on the present value of $x[n]$.
static

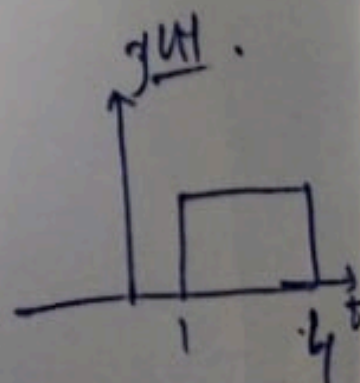
yes absolutely Summable
stable.

$y[n] = \{ \underset{\uparrow}{2}, 0, -5, 1 \}$

(iii) $y_{CH} = x(t-1) - x(t-4)$ $x_{CH} = u_{CH}$
~~not~~ static.

$y_{CH} = u(t-1) - u(t-4)$

Then $y_{CH} = \underline{\underline{stable}}$



x_{CH} is banded